INTRODUCTION

- What is a data?

  - It is a set of values of **qualitative variables**
    - Transcripts of interviews or focus group. These can be used to capture people's perspectives, opinions, and experiences on a particular topic. ...
    - Field notes. ...
    - Case studies. ...
    - Open-ended survey responses. ...
    - Document analysis. ...
    - Observation note. ...
    - Audio or video recordings
  - and **quantitative variables**
    - Weight in pounds
    - Length in inches
    - Distance in miles
  - It is measured, collected and reported and analyzed

- The need of the hour is to understand, manage, process and take the data for analysis to draw valuable insights

- What is a database?
  - A **database is a collection of data**
  - **describes the** activities of one or more related organizations.

- What is a database management systems?
  - A **Database Management System or DBMS is software** designed to assist in maintaining and utilizing large collections of data.
  - Program to access the data(in simple terms)

Database Systems

- DBMS contains information about a particular enterprise
  - Collection of interrelated data
  - Set of programs to access the data
  - An environment that is both *convenient* and *efficient* to use
- Database systems are used to manage collections of data that are:
  - Highly valuable
  - Relatively large
  - Accessed by multiple users and applications, often at the same time.
- A modern database system is a complex software system whose task is to manage a large, complex collection of data.
- Databases touch all aspects of our lives

Database-System Applications

- Enterprise Information
  - Sales: customers, products, purchases
  - Accounting: payments, receipts, assets
  - Human Resources: Information about employees, salaries, payroll taxes.
- Manufacturing: management of production, inventory, orders, supply chain.
- Banking and finance
  - customer information, accounts, loans, and banking transactions.
  - Credit card transactions
  - Finance: sales and purchases of financial instruments (e.g., stocks and bonds; storing real-time market data
- Universities: registration, grades

Database-System Applications

- Airlines: reservations, schedules
- Telecommunication: records of calls, texts, and data usage, generating monthly bills, maintaining balances on prepaid calling cards
- Web-based services
  - Online retailers: order tracking, customized recommendations
  - Online advertisements
- Document databases
- Navigation systems: For maintaining the locations of varies places of interest along with the exact routes of roads, train systems, buses, etc.

Purpose of Database Systems / Drawbacks of File System

In the early days, database applications were built directly on top of **file systems,** which leads to:

- Data redundancy and inconsistency: data is stored **in multiple file formats** resulting induplication of information in different files
- Difficulty in accessing data
  - Need to write a new program to carry out each new task
- Data isolation
  - Multiple files and formats
- Integrity problems
  - **Integrity constraints** (e.g., account balance > 0) become “buried” in program code rather than being stated explicitly
  - Hard to add new constraints or change existing ones

Purpose of Database Systems /Drawbacks of File system

- Atomicity of updates
  - Failures may leave database in an **inconsistent state** with partial updates carried out
  - Example: **Transfer of funds from one account to another should either complete or not happen at all**
- Concurrent access by multiple users
  - Concurrent access needed for performance
  - Uncontrolled concurrent accesses can lead to **inconsistencies**
    - Ex: Two people reading a balance (say 100) and updating it by withdrawing money (say 50 each) at the same time
- Security problems
  - Hard to provide **user access to some, but not all, data**

Database systems offer solutions to all the above problems

University Database Example

- Use university database to illustrate all the concepts
- Data consists of information about:
  - Students
  - Instructors
  - Classes
- Application program examples:
  - Add new students, instructors, and courses
  - Register students for courses, and generate class rosters
  - Assign grades to students, compute grade point averages (GPA) and generate transcripts